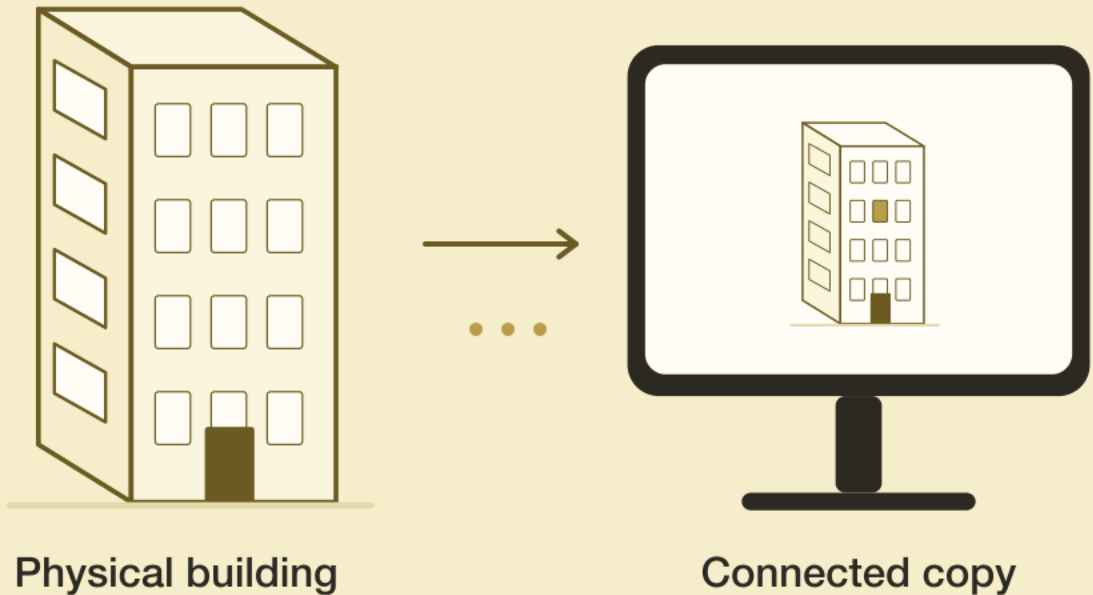


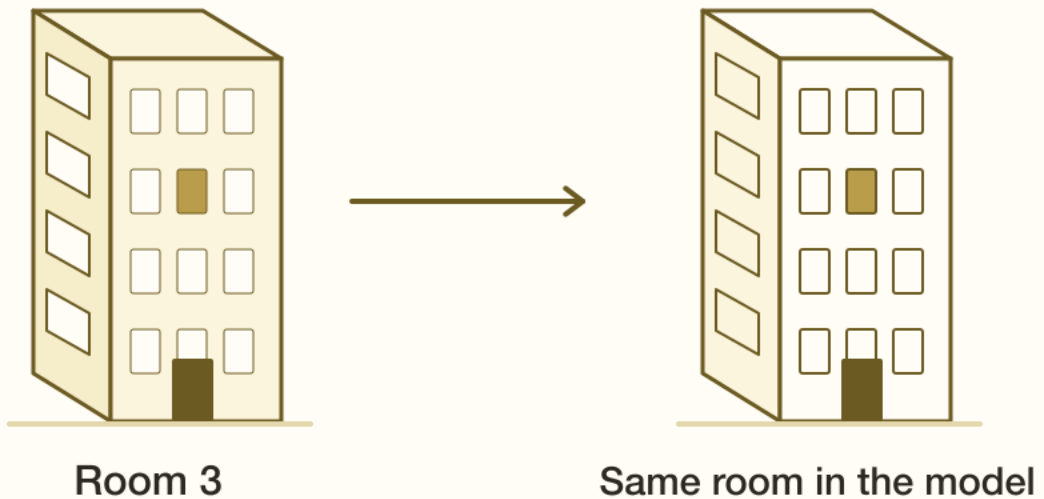
A building. A second life on a screen.

How can a building have
a live digital copy?



Start with the real building.

Then give its rooms and equipment a place in a digital model.



The model gives the physical room a digital identity.

Sensors give it something to read.

Small devices measure what is happening.
The model can receive those readings.



ROOM TEMPERATURE

24°C

Room 3



ENERGY TODAY

18 kWh

Building



WATER SENSOR

Dry

Utility room

Illustrative readings. No actual building data.

A number needs an address.

24°C means more when you know it belongs to Room 3, on Floor 2.



Sensor A



Room 3

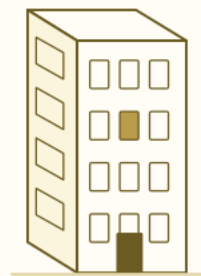


Floor 2



Building

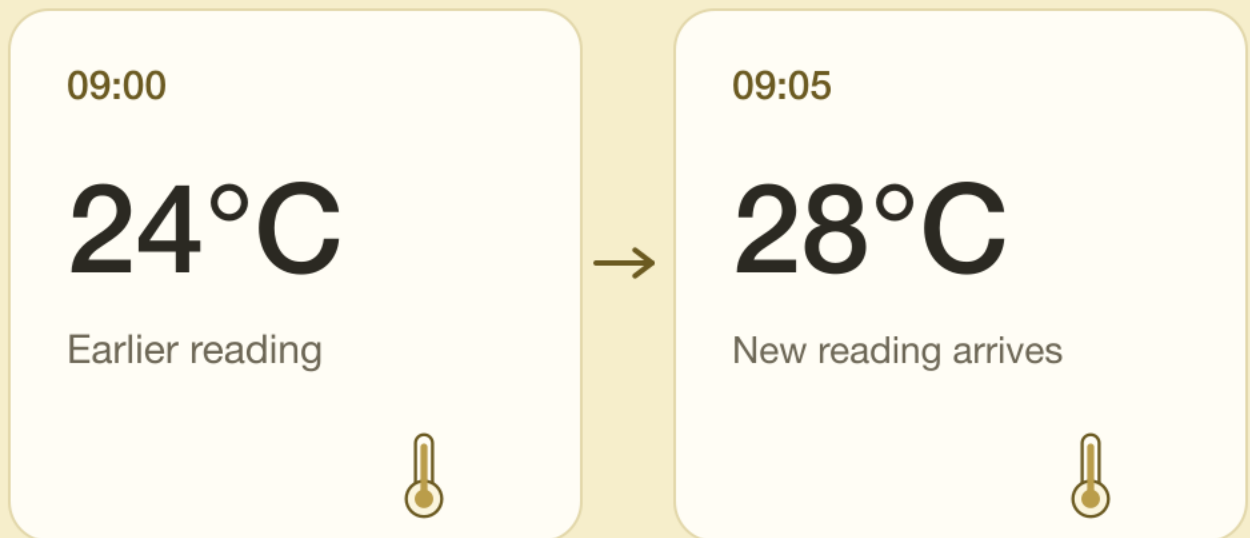
24°C



Illustrative room and sensor relationships.

New reading. Updated copy.

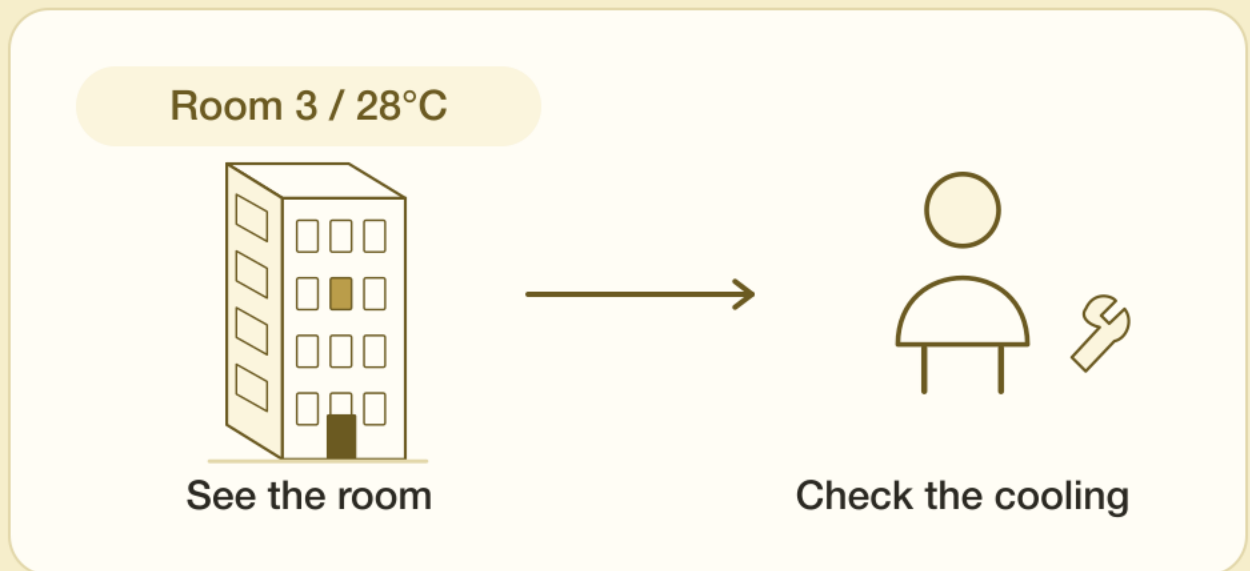
As new data arrives, the model changes.
Update speed depends on the system.



Illustrative values and timing.
A connected model is not necessarily instant.

Now a person can act on it.

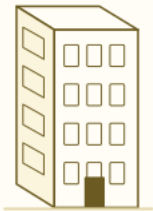
A room gets unusually warm.
The manager knows where to start checking.



Illustrative example. A warm room is a clue,
not proof that a particular part has failed.

A 3D tour shows what it looks like.

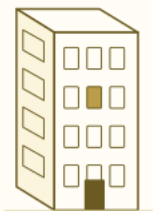
A connected twin can also show what is happening inside it.



STATIC 3D TOUR

Shape and layout

No live building data required



CONNECTED TWIN

Shape + operational data

Conditions can change with new data

A beautiful model alone does not make a live twin.

Some twins can test ‘what if?’

Could a different cooling schedule help?
A simulation estimates the result first.

FICTIONAL COOLING SCHEDULE

Current



Proposed



Run a simulation before changing the building.

Model estimate. Not a guaranteed result.

The copy has a blind spot.

If a sensor stops sending, a reading may be old. Check when it last updated.

● SENSOR OFFLINE

24°C

Old reading

Last updated 09:00

Illustrative example. An old value can still look precise.
It does not tell you the temperature now.

Real building. Connected data. Better questions.

That is the idea of a digital twin.
It helps people know what to check next.



A clearer view of the building, not a crystal ball.